Darshan Summary Report

Job Summary

Job ID	1347640
User ID	8391
# Processes	1
Run time (s)	298.6351
Start Time	2026-02-26 19:46:52
End Time	2026-02-26 19:51:51
Command Line	python3 /lcrc/project/ATLAS-HEP-group/rwang/Argonne_computing/PPS-CCE/SML/ML_Darshan/gsoc26/PyTorch_Fash --data_path /lcrc/project/ATLAS-HEP-group/rwang/Argonne_computing/PPS-CCE/SML/ML_Darshan/gsoc26/P --threads 2 --batch_size 64 --epochs 10

Darshan Log Information

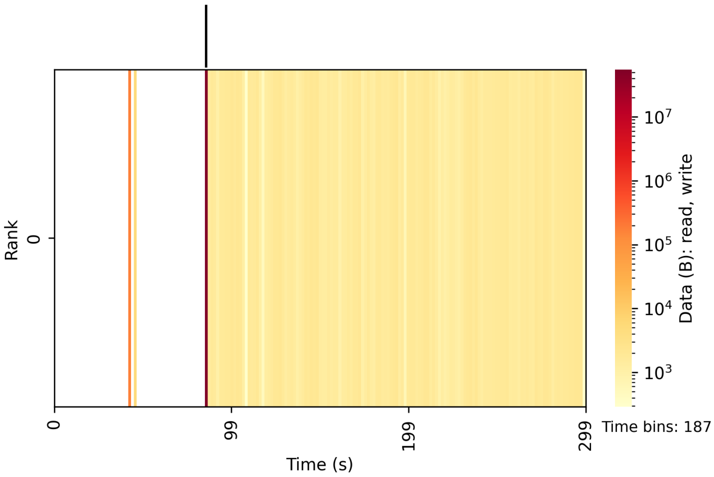
Log Filename	rwang_python3_id1347640-1347640_2-26-49612-2664695618105195845_1.darshan
Runtime Library Version	3.5.0
Log Format Version	3.41
POSIX (ver=4) Module Data	0.70 KiB
STDIO (ver=2) Module Data	0.07 KiB
DXT_POSIX (ver=2) Module Data	29.81 KiB
HEATMAP (ver=1) Module Data	0.45 KiB

I/O Summary

Heat Map: HEATMAP POSIX

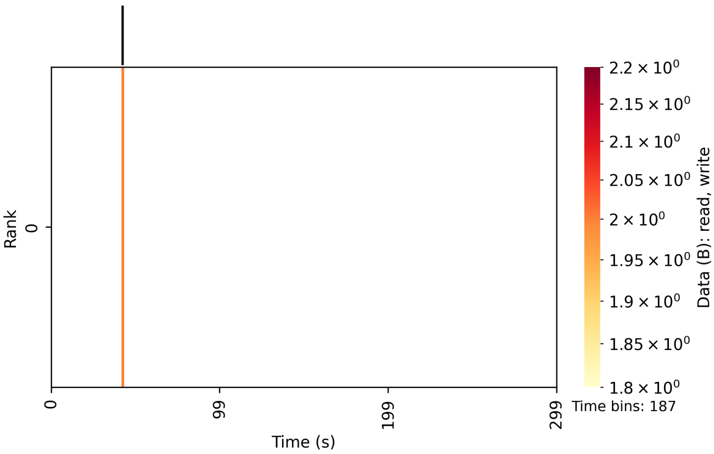
Heat Map: DXT_POSIX

DXT_POSIX trace data available, but DXT-based heatmaps are not currently generated by default by the job summary tool. To force generation of heatmaps based on DXT trace data, users can specify the '--enable_dxt_heatmap' option.



Heat map of I/O (in bytes) over time broken down by MPI rank. Bins are populated based on the number of bytes read/written in the given time interval. The top edge bar graph sums each time slice across ranks to show aggregate I/O volume over time, while the right edge bar graph sums each rank across time slices to show I/O distribution across ranks.

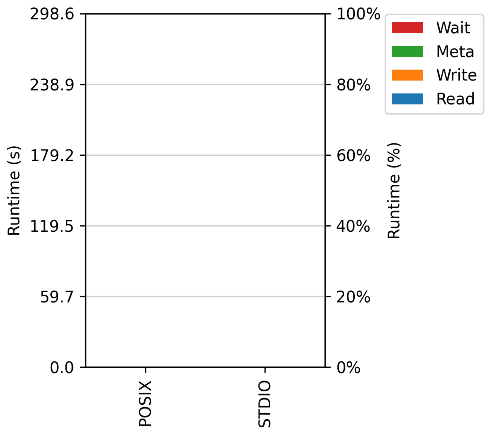
Heat Map: HEATMAP STDIO



Heat map of I/O (in bytes) over time broken down by MPI rank. Bins are populated based on the number of bytes read/written in the given time interval. The top edge bar graph sums each time slice across ranks to show aggregate I/O volume over time, while the right edge bar graph sums each rank across time slices to show I/O distribution across ranks.

Cross-Module Comparisons

I/O Cost



Average (across all ranks) amount of run time that each process spent performing I/O, broken down by access type. See the right edge bar graph on heat maps in preceding section to indicate if I/O activity was balanced across processes. The 'Wait' category is only meaningful for PNETCDF asynchronous I/O operations.

Per-Module Statistics: POSIX

Overview

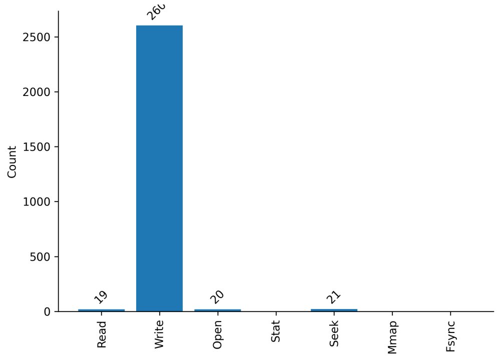
files accessed	**8**
bytes read	**52.58 MiB**
bytes written	**213.37 KiB**
I/O performance estimate	**242.81 MiB/s (average)**

File Count Summary (estimated by POSIX I/O access offsets)

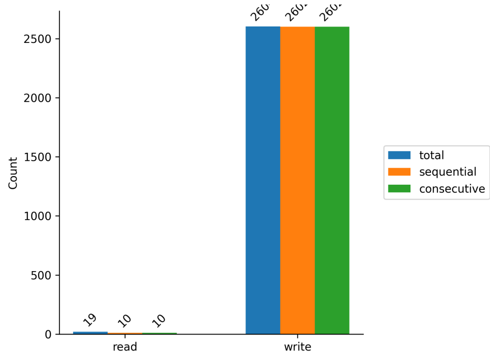
	number of files	**avg. size**	**max size**
total files	8	6.59 MiB	44.86 MiB
read-only files	5	10.50 MiB	44.86 MiB
write-only files	2	106.68 KiB	212.81 KiB
read/write files	0	0	0

Operation Counts

Access Pattern

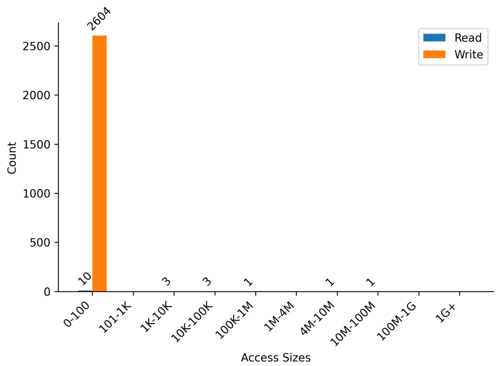


Histogram of I/O operation frequency.



Sequential (offset greater than previous offset) vs. consecutive (offset immediately following previous offset) file operations. Note that, by definition, the sequential operations are inclusive of consecutive operations.

Access Sizes



Histogram of read and write access sizes. The specific values of the most frequently occurring access sizes can be found in the *Common Access Sizes* table.

Common Access Sizes

Access Size	Count
84	235
88	231
78	230
90	228

Per-Module Statistics: STDIO

Overview

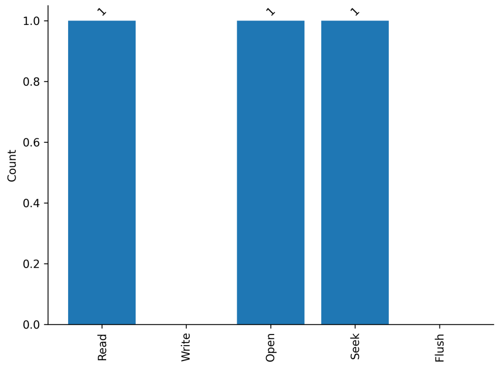
files accessed	**1**
bytes read	**2 Bytes**
bytes written	**0 Bytes**
I/O performance estimate	**0.01 MiB/s (average)**

File Count Summary (estimated by STDIO I/O access offsets)

	number of files	avg. size	max size
total files	1	2 Bytes	2 Bytes
read-only files	1	2 Bytes	2 Bytes

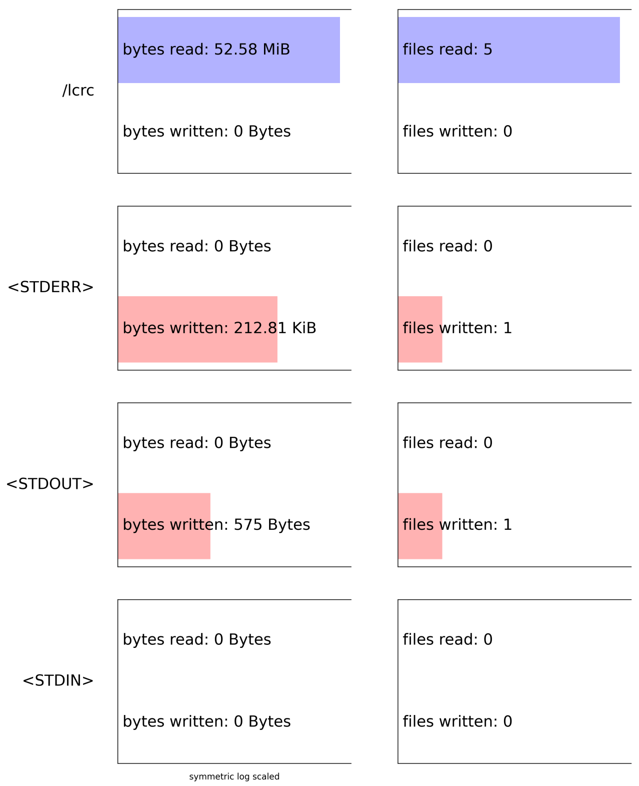
write-only files	0	0	0
read/write files	0	0	0

Operation Counts



Histogram of I/O operation frequency.

Data Access by Category



Summary of data access volume categorized by storage target (e.g., file system mount point) and sorted by volume.

For more information visit the Darshan website